Galaxy Morphology Classification

Alexandre Gauthier,* Archa Jain,† and Emil Noordeh‡
Stanford University
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We apply machine learning techniques to the problem of galaxy morphology classification. We use both supervised and unsupervised methods to study the Galaxy Zoo dataset of 61,578 pre-classified galaxies. After image pre-processing, we perform multi-class classification using the following algorithms: one-vs-all SVM with an RBF kernel, decision tree, random forest, AdaBoost classifier, and KNN. We classify galaxies as either spiral, elliptical, round, disk, or other and find the best performing algorithm to be random forest with 67% classification accuracy. We also use regression to predict the probabilities of galaxies being associated with each class with 94% accuracy. We visualize the dataset using K-means clustering and agglomerative clustering unsupervised learning algorithms. The principal sources of variation between galaxy images are revealed to be correlated with brightness and eccentricity.

I. INTRODUCTION

There are hundreds of billions of galaxies scattered throughout the cosmos which define the structure of the universe on the largest scales. The distribution of the physical properties of these galaxies hold the clues necessary for our understanding of the past, present, and future of the universe. Astronomers and cosmologists rely on large scale, observational studies of these properties to inform their theoretical models and provide valuable insight into new science.

Over the past few decades, advances in technology have allowed large scale surveys of the sky to catalog galactic data at unprecedented rates. The Sloan Digital Sky Survey (SDSS) alone has cataloged over 1.2 billion objects across one third of the sky [1]. Not only is this a testament to the vastness and complexity of our universe but also to the need for robust, automated analysis techniques to interpret such immense amounts of data. These techniques need to be refined rapidly as next generation surveys, such as the Large Synoptic Survey Telescope (LSST), will add tens of billions of new objects to these catalogs within the next decade [2].

One of the primary ways astronomers extract physical information from the photometry of a galaxy is by looking at its morphology. For instance, spiral galaxies contain cold gas feeding young stellar nurseries while elliptical galaxies contain primarily old and dying stars. Astronomers classify galaxies into morphological categories; the distribution of morphologies across spacetime tells us about the evolution of the cosmos. One such classification scheme is the “Hubble tuning-fork” shown in fig. 1. The tuning-fork can be used to broadly classify galaxies into elliptical, spiral, or irregular (other) morphologies.

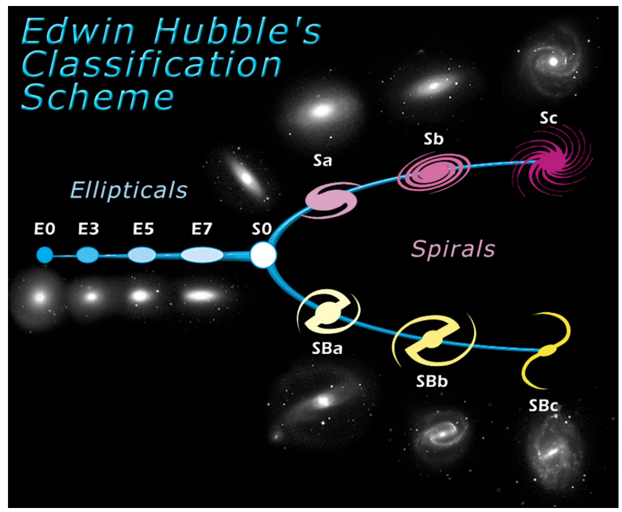

FIG. 1. “Hubble tuning-fork” galaxy classification scheme [ESA/Hubble].

A. Galaxy Zoo Project

Classifying galaxies into these and other morphological classes is a task that humans excel at. To that end, the Galaxy Zoo (GZ) project [3] was launched to crowdsource the classification of a large sample of SDSS galaxies. GZ is a web-based program where citizen scientists can classify galaxies by following a programmed decision tree. To date, over 100,000 members have classified nearly 1 million objects by hand. This data has been corrected for multiple biases and shown to be in strong agreement with classifications made by professional astronomers [4]. The GZ dataset has already fueled a number of remarkable papers in astronomy [4–6], and provides an invaluable training set for automated classification algorithms that could perhaps classify billions of more objects. In our project, we use the GZ dataset to train and test a number of machine learning classifiers. We also examine the inherent structure of the data using unsupervised learning.

* agau@stanford.edu
† archa@stanford.edu
‡ emiln@stanford.edu

B. Related Work

The use of machine learning to classify galaxies has been an active area of research over the past two decades. Some of the early work in the field attempted to use neural networks, decision trees, and Naïve-Bayes classifiers trained on relatively small datasets of hundreds of objects [7–9]. At best, these attempts achieved classification errors of ∼20%. Recently, more advanced techniques have been successfully applied to larger datasets, including the GZ dataset. A neural network was trained on 900,000 objects from GZ using a novel set of features and achieved over 90% classification accuracy [10]. Another group used convolutional neural nets to classify a subset of the 50,000 brightest objects in the GZ dataset with over 99% accuracy [11].

While these results are impressive, the vast majority of classifiers in the literature classified objects into roughly three bins: elliptical galaxies, spiral galaxies, and other. In order to take full advantage of the science contained in the SDSS, LSST, and other surveys, more detailed object classifications need to be made. Prior attempts at extending analyses in this way have performed extremely poorly. For instance, Calleja *et al.* (2004) found that just going from 3 to 5 categories dropped the accuracy of their neural network from 90% to 50% [12]. While achieving accurate sub-classifications at the level of fig. 1 is still quite far away, in this paper we extract additional information from GZ images such as the "roundness" and "diskiness" of the galaxies. This is a step in the right direction for galaxy classification algorithms.

C. Dataset Description

The GZ data we used comes in two parts, with $61,578$ pre-classified galaxies provided. The "X" data is made up of 424×424 pixel 3-channel RGB images of galaxies taken from SDSS. GZ also provides the classifications that have been collected from the crowdsourced quiz (the "Y" data). The classifications are given in the form of probabilities that quiz respondents answer a certain way to 37 questions. Because some questions only appear if earlier questions are answered a specific way, the probabilities for different answers to the same question do not always add up to one.

To convert the 37 GZ questions into useful galaxy classes (e.g. "elliptical"), we performed a comparative analysis of the provided labels to assign each galaxy to one of five categories: disc, spiral, elliptical, round, and other.

We assign a galaxy to "other" if a plurality of respondents answer in the first question that an image contains a star or artifact, if most people say there's "something odd," or if it's not in one of the other categories. If a plurality answer in the first question that the galaxy is smooth, we place that galaxy into either the elliptical or the round category, depending on how people answered the second question. Finally, if a plurality of survey takers answer in the first question that there are "features," we assign the galaxy to either disc or spiral categories, depending on the answers to subsequent questions relating to the spiral or disc-like nature of the galaxy.

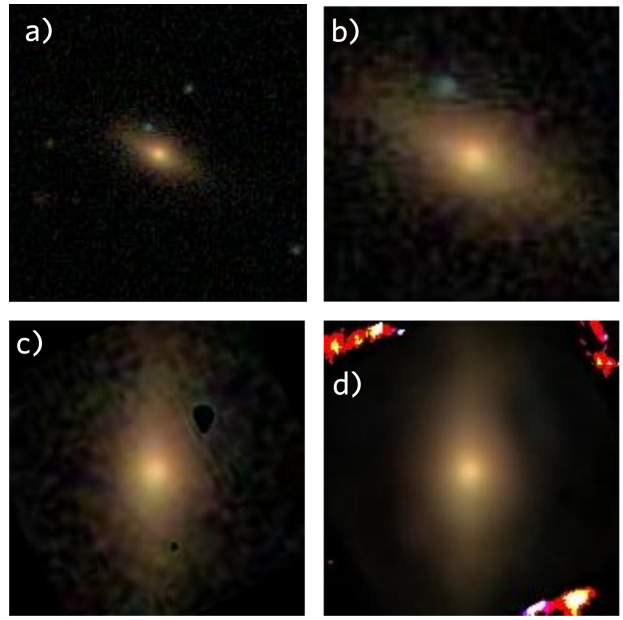

FIG. 2. Illustration of our image pre-processing procedure. The original galaxy image in (a) is first cropped to the size of the primary galaxy (b). We then rotate to align the principal axis of the galaxy vertically, and subtract background objects (c). Finally, we apply a 125-component PCA algorithm. The reconstructed image after PCA is shown in (d).

II. IMAGE PRE-PROCESSING

A. Cropping, rotation, and background subtraction

Images in the GZ dataset consist of large fields of view with the galaxy of interest in the center. They often contain secondary objects (sometimes overlapping with the primary object); a significant amount of pre-processing is necessary before we can apply machine learning algorithms. Our image pre-processing pipeline is illustrated in fig. 2.

We start by cropping the images to reduce the file size and remove many secondary objects. We then rotate the galaxies such that their principal axes are vertical. Next we subtract the background by carving out isophotes at the 10% maximum intensity threshold in each image and setting the pixel values in all non-central isophotes to zero.

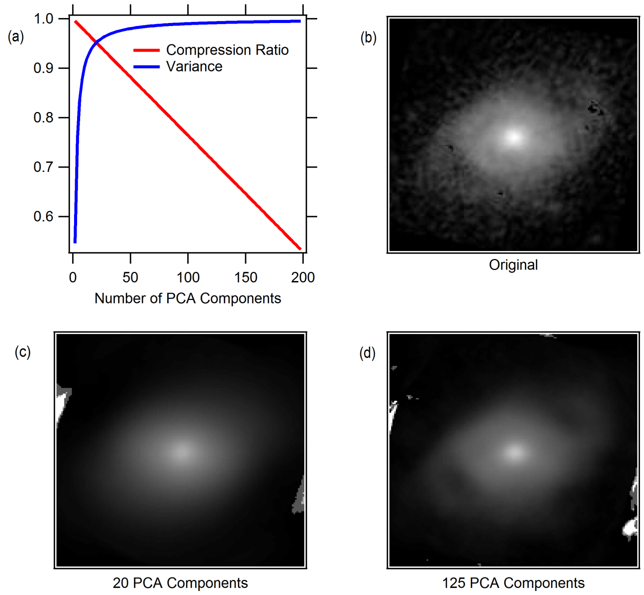

FIG. 3. The performance of the PCA model as a function of number of features is plotted in (a). We trained PCA models on 5,000 samples. We reconstructed the galaxy shown in (b) for a 20-feature PCA model (c) as well as for a 125-feature model (d). The 20-feature model fails to preserve the spiral arms.

B. Principal Component Analysis

Since the images are fairly large, if we were to work on the raw pixel data, we would have 76,800 features per image. Training such a model on the whole dataset would be computationally-intensive and would likely lead to overfitting. Since most of the images are fairly similar - a small, bright galaxy in the center with a dark background - we expect many of these features to be highly-correlated. We take advantage of this by using principal component analysis (PCA) to reduce the number of features in our data.

Figure 3 shows an analysis of algorithm performance as a function of PCA features. Although 95% of the variance in the data can be explained using only 20 PCA features, this level of compression destroys a lot of the galactic information we're looking for. Instead, we use 125 PCA features, which preserves > 99% of the variance in the data.

III. RESULTS

A. Supervised learning

For our classification work we classify galaxies into one of five categories: disc, spiral, elliptical, round, and other. These categories are described in section I C.

We source our machine learning algorithms from the Scikit-learn [14] Python package. We performed multi-class classification using a variety of techniques. We applied a one-vs-all classifier using an SVM with an RBF kernel, decision tree, AdaBoost classifier, K-nearest-neighbors and random forest. The classification accuracies for these algorithms as determined using 10-fold cross validation (CV) are shown in table I. The best performing algorithm was the random forest with 67% accuracy. The random forest confusion matrix for a subsample of our data is shown in table II.

TABLE I. Classifier Accuracy

Classifier	Training Accuracy	CV Accuracy
Random Forest	0.75	0.67
AdaBoost	0.77	0.62
KNN	0.60	0.57
Decision Tree	0.91	0.53
SVM	0.49	0.43

TABLE II. Random Forest Confusion Matrix

True:	Disc	Spiral	Elliptical	Round	Other
Predicted Disc	**620**	69	102	3	56
Predicted Spiral	35	**1524**	156	83	184
Predicted Elliptical	48	213	**1299**	158	143
Predicted Round	0	131	149	**1241**	67
Predicted Other	48	790	366	212	**303**

A random forest is a classifier consisting of a collection of treestructured classifiers $\{h(x, \Theta_k), k = 1, ...\}$ where the $\{\Theta_k\}$ are independent identically distributed random vectors and each tree casts a unit vote for the most popular class at input x [13]. It is similar to Adaboost except that the random forest picks each successive "tree" randomly (in contrast, Adaboost has no random elements and grows an ensemble of trees by successive reweightings of the training set where the current weights depend on the past history of the ensemble formation).

The primary source of error in all of our classification algorithms was the mis-classification of spiral galaxies into the "other" category as can be seen in table II. This is due to a large number of "spiral" sources having very faint spiral features, making them difficult to classify even for humans. For such galaxies, survey takers were divided about how to answer the question "Is there any sign of a spiral arm pattern?" One of the spiral galaxies on which our random forest algorithm fails is shown in fig. 4. Clearly the spiral structure is difficult to see in this image and it is not surprising that human classifiers are split on the appropriate classification.

Since human classifications are inconclusive for a number of the galaxies, we decided that a regression approach to the problem might be more suitable. Instead of performing multi-class classification, we attempted to directly predict the probability distribution function (PDF) of the classifications, allowing our models to account for those galaxies with uncertain classification.

FIG. 4. The spiral arms are hard to see in this galaxy, causing the associated spiral probability to be close to 0.5.

TABLE III. Decision tree regression results

Category	Training Accuracy	CV Accuracy
Spiral	0.95	0.93
Ellipse	0.93	0.93
Disc	0.99	0.96
Round	0.96	0.95
Other	0.96	0.94

We do this using a random forest regressor with 200 estimators. We measure the accuracy A_k of our regressor on class k as

$$A_k = \frac{1}{m}\sum_{i=1}^{m}\sum_{j=1}^{5}\mathbf{1}\{j=k\}|P^j_{i,true} - P^j_{i,pred}|$$

where m is our sample size, and $P^j_{i,true/pred}$ are the true and predicted probabilities for class j of galaxy i. We find an average accuracy over all classes of 0.95. The detailed regression results for each class are given in table III. Given that in reality galaxies lie on a continuum between different classes, predicting the PDF of each galaxy instead of forcing a classification can potentially provide more valuable insight into its morphology.

B. Unsupervised Learning

The GZ questionnaire is set up as a decision tree, with each question the respondent answers leading to a different question. Since we are only classifying into 5 categories, we use clustering to explore the inherent structure of the entirety of the dataset. This also allows us to ascertain how well the structure of the data aligns with the GZ decision tree.

We apply three clustering algorithms to the 125-component PCA dataset - k-means, and two hierarchical algorithms (agglomerative with different linkage schemes) with varying results. To evaluate the clusters, we used the silhouette score to quantify "tightness" and separation of each resulting cluster (fig. 5). Agglomerative clustering with complete linkage displayed slightly higher silhouette scores than the other two algorithms, although it failed for 2 clusters. It is also interesting to note that agglomerative clustering with Ward linkage performed significantly worse. This suggests that most of the data is fairly overlapping, producing cleaner results with a stricter linkage metric.

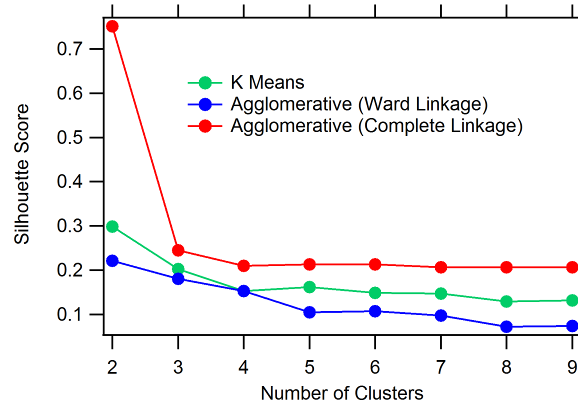

FIG. 5. Comparison of clustering algorithms. The peak at $n = 2$ for agglomerative clustering with complete linkage is anomalous as it places all but one galaxy into the same cluster.

The agglomerative clustering algorithm with complete linkage is slow to run. This led us to use the second-best k-means clustering for further analysis; this choice allows us to achieve an order of magnitude shorter runtime. The k-means clustering algorithm works by minimizing

$$\sum_i |x^{(i)} - \mu_{c_i}|^2$$

with respect to $\boldsymbol{\mu}$, where μ_α is the center of the αth cluster, and point $x^{(i)}$ is assigned to cluster c_i.

We computed the correlation between clustering and each of the 37 GZ response probabilities. There is little correlation ($r < 0.3$) for most responses. Questions related to whether a galaxy is rounded or elliptical (eccentricity) have a higher $r \sim 0.6$.

In order to visualize the distribution of clusters, we performed another round of PCA compression on the 125-dimensional data in order to reduce it to two dimensions. We then plotted the galaxies on a two-dimensional scatter plot. The clusters of galaxies we found using the 125-dimensional data are distinctly separated in the reduced two-dimensional space for up to 6 clusters (fig. 6b). This indicates that the much of the variation in the data comes from these two principal components.

Since we only see correlation between clusters and questions relating to galactic eccentricity, we tried plotting our galaxies in the reduced two-dimensional space with a color scale indicating eccentricity as determined by Galaxy Zoo survey respondents (fig. 6d). This shows a pattern revealing that one of the two principal com-

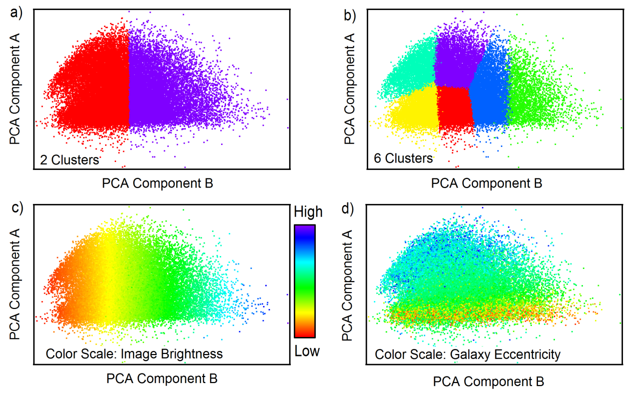

FIG. 6. Galaxies plotted in the space of their two principal components. Colors in (a) and (b) represent cluster found using k-means clustering for 2 and 6 clusters, respectively. (c) colors galaxies by brightness; (d) colors galaxies by eccentricity as determined by Galaxy Zoo survey takers.

ponents is correlated with galactic eccentricity. We also found that the other principal component is extremely strongly correlated with the brightness of a galaxy as measured by the sum of pixel values across the image (fig. 6c).

The spiral labels provided by galaxy zoo display low correlation ($r < 0.2$) with the clusters we found. This helps explain why we had difficulty discerning spiral galaxies using classification algorithms. The principal variation between galaxies is contained in brightness and shape rather than the presence of internal structure such as spiral arms.

The visualization and the earlier analysis with the Silhouette scores suggests that the data lies on a fairly continuous spectrum, without any clearly separable clusters. This is consistent with our earlier findings with supervised learning where the classifiers performed poorly because of galaxies that are on the "edge" of two classes. This also helped inform our move to framing the supervised learning as a regression problem.

IV. SUMMARY

In summary, we have applied supervised and unsupervised learning approaches to classifying galaxy morphologies. We found that direct classification of galaxies into 5 groups worked quite poorly with our best performing algorithm, the random forest, achieving only 67% classification accuracy. This was consistent with our findings from unsupervised learning that indicated that the data is fairly continuous and not easily separable. We had more success with a regression approach attempting to estimate a PDF for each galaxy; this approach achieved 95% accuracy using a random forest regressor.

Unsupervised analysis revealed that brightness and eccentricity contain a significant fraction of the variation between galaxies. Normalizing images by brightness could help extract additional information on more relevant morphological features.

It would be interesting to apply our regression based approach to determining galaxy morphologies to more detailed morphological sub-classes as illustrated in fig. 1. Attempts in the literature at doing such classifications have always performed very poorly since many galaxies fall into an intermediate region between two different morphological classes. Estimating the PDF instead of directly classifying objects is more robust to this behavior and may be a better approach to take.

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